Alberta Palaeontological Society

The Driftwood Canyon Fossil Beds, Smithers, Northern British Columbia – An Amazing Well-Preserved and Highly Diverse Eocene-Age Terrestrial Paleoecology

Speaker: Tako Koning, P. Geol., Senior Geologist - Consultant
Location: Room B108, Mount Royal University, Calgary, AB
November 20, 2026, 7:30 PM – 9:00 PM.

A Famous Site for Eocene Fossils

Driftwood Canyon Provincial Park is located 10 km northeast of Smithers, BC in the Bulkley River valley and contains one of the richest fossils beds in the province of British Colombia. In the field of paleontology, it is recognized as one of the world's most significant fossil beds due to its 52-million-year-old Eocene-age fossils-rich shale called the Driftwood Shale. The site offers easy access on foot to view remarkably well-preserved fossils of terrestrial and aquatic plants, insects, rare fish fossils and early mammals. It is ranked as a premier location for studying Eocene warm-climate ecosystems in northern BC.

Climatic Conditions

The finely laminated shale beds in the Driftwood Canyon record an ancient freshwater swamp or lake environment where fine sediments were trapped. The formation of the lake now exposed by the Driftwood Canyon approximately coincided with the onset of basaltic volcanic activity in this area about 51 million years ago. The climate back in the Eocene in the location of the Driftwood Canyon was very different from that of today. The earth was much hotter, to the point that there was likely no permanent ice at either of the poles. Atmospheric carbon dioxide levels are estimated to have been much higher than they are today.

Flora

The flora growing around the Driftwood Canyon in the Eocene was a mix of familiar and exotic species. Fossils of pine, alder, birch, cypress, elm, ferns, fir, hemlock, oak, pines, sassafras and spruce have been found but the more common plant species is the *Metasequoia* or the dawn redwood. Also found has been fossils of ginko and *Tetracentron* - also of *Dipteronia*, a type of soapberry.

Figure 1. A "peel" of an *Eosalmo driftwoodensis* fish fossil from the Bulkley Valley Museum, Smithers, BC. September 15, 2020 museum article.

Figure 2. *Metasequoia,* length 5 cm.

Fauna

Fossilized insects are also common in the Driftwood shale beds. Water striders are some of the most well-preserved, with their distinctive spindly legs, but dozens of other species have been found including aphids, march flies, mosquitos, gnats, ants, termites, weevils and wasps. Also discovered have been beetle fossils which were obligate palm-feeders inferring that there must have been palm trees growing at the site to sustain these populations. One of the more ubiquitous fossils is *Eosalmo driftwoodensis* or the freshwater dawn salmon. Other fish fossils discovered include suckerfish, trout, and bowfins, a type of air breathing fish. Bird feathers are infrequently collected from the shales. The first discovery of a bird fossil was in 1968 and was identified as a puff bird *Picformes bucconidae*. The second bird was collected in 1970 and is a long-legged water bird. More recent studies in 2019 have identified both birds as coliform birds and as members of the Songzidae respectively. Only two fossils of mammals have been discovered. In 2004, paleontologists discovered jaws of *Heptodon* an early relative of the tapir. Also discovered was a tiny hedgehog that was estimated to be only 5 cm long. This news species was called *Silvacola acares,* meaning “little forest dweller”.

Biography

Tako Koning is Holland-born and Canada-raised with a B.Sc. Geology in 1971 from the University of Alberta and a B.A. in Economics in 1981 from the University of Calgary. He has worked for fifty years in oil and gas exploration and development as a geologist, manager and executive including vice president exploration with Texaco Canada Petroleum Inc. Approximately twenty years of his working life has been in Calgary. He and his wife Henrietta also lived and worked for thirty years in Indonesia, Nigeria and Angola. Since the age of ten (or younger), he has had an abiding interest in fossils and the fields of geology and paleontology. This has led to him being an active member of the Canadian Energy Geoscience Association (CEGA) and the Alberta Palaeontological Society (APS) leading field trips every summer for both organizations to see the K/Pg mass extinction boundary at Knudsen’s Farm near Huxley, Alberta. Also, for CEGA and the APS, he leads an annual half day Tyndall Stone Tour to view 450-million-years old Ordovician-age fossils-rich decorative limestones in inner city Calgary.